

Zhuoyuan (Joey) YU

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Research interests: Long-Horizon Assembly, Robot Skill Learning, Multi-Agent Systems

EDUCATION

University of Auckland		Auckland, New Zealand
Major: Doctor of Philosophy in Mechatronics Engineering		08/2025-Present
- Research focus: Robot Skill Learning for Long-Horizon Assembly - Scholarship: MBIE Scholarship (Full Funding)		
National University of Singapore	GPA: 5.0 / 5.0	Singapore
Major: M.Eng. in Mechanical Engineering (By Research)		08/2023-05/2025
Research focus: Robotics, Multi-agent Deep Reinforcement Learning, Control		
Northwestern Polytechnical University	GPA: 83.1%	Xi'an, China
Major: B.Eng. in Aircraft Design and Engineering		09/2019-07/2023
- Scholarship: Second Prize Scholarship - NPU School of Aeronautics Student Union President - International Internet+ College Students Innovation and Entrepreneurship Competition "Huamo Cup" National College Students Flight Simulation Championship	Excellent Student Leader 04/2021-06/2022 National Gold Award National Third Prize	

RESEARCH EXPERIENCE

➤ Robot Skill Learning for Long-Horizon Assembly	08/2025-Present
[Industrial AI Group] Supervisors: Lu Yuqian, Andrew McDaid and Bruce MacDonald	
Enhance the precision of long-horizon assembly tasks.	
Related: Python, Pytorch, ROS1, Robot Skill Learning, Vision Language Action	
➤ Multi-Agent Path Finding Based on Deep Reinforcement Learning	05/2024-01/2025
[Joint Program between NUS and A*STAR] Supervisors: Guo Hongliang and Chew Chee Meng	
- Improved the existing Node2Vec algorithm to handle the dynamic topological networks better. - Utilized Graph Attention Networks to enhance the decision-making weights of dynamic edges. - Integrated reinforcement learning for online training of the network.	
Related: Python, Pytorch, ROS1, Multi-Agent Systems, Graph Attention Networks, Natural Language Processing	
➤ Design and Control of Manta Ray Robot (Bioinspired Underwater Robot)	08/2023-05/2025
[NUS ME Control and Mechatronics Lab] Supervisor: Chew Chee Meng	
- Designed a new type of buoyancy system and mass adjustment system for the Manta Ray robot. - Improved original single-degree-of-freedom pectoral fin to dual-degree-of-freedom, enhancing controllability. - Upgraded Arduino-based control system to include control of the buoyancy system and the pectoral fins.	
Related: Python, Arduino, SolidWorks, Bioinspired Robotics	
➤ Quadcopter Overall Design and Trajectory Re-planning	07/2020-09/2021
[NPU Aircraft Design and Testing Technique Engineering Laboratory] Supervisor: Wang Ban	
- Designed and made a quadcopter unmanned aerial vehicle (UAV). - Studied the trajectory re-planning and obstacle avoidance of UAVs.	
Related: MATLAB, XFLR5, Catia, Aerodynamics	

SELECTED PROJECTS

➤ Autonomous Mobile Robotics [GitHub] ROS1, Python, OpenCV	01/2024-04/2024
Integrate vision recognition to navigate the robot in a simulation environment and search objects.	
➤ Manipulator Simulation [GitHub] MATLAB, Kinematics	03/2024-04/2024
Establish a robotic arm model, then use inverse kinematics to calculate the corresponding joint angles.	

PUBLICATIONS (* corresponding author)

Yu Zhuoyuan., Guo, H*. , Chew, C. M, Adiwahono, A. H., Chan, J., Tynn, B. S. W., & Yau, W. Y, “[Multi-Robot Reliable Navigation in Uncertain Topological Environments with Graph Attention Networks](#),” *IEEE Robotics and Automation Letters* (2025)

TECHNICAL SKILLS

Programming Language: Python, MATLAB, C/C++

Software: Pytorch, ROS1, Gazebo, SolidWorks, Catia, OpenCV, LaTeX, Git, Isaac Lab

Hardware: Arduino, Mechanical Design, 3D Printing